Session 1 -

1. Name 5 IT job roles and write one line about what each role does.

1. **Software Developer** – Designs, writes, and maintains software applications.
2. **Web Developer** – Builds and maintains websites and web applications.
3. **Software Tester (QA Engineer)** – Tests software to find bugs and ensure quality.
4. **System Administrator** – Manages and maintains computer systems, servers, and networks.
5. **Database Administrator (DBA)** – Manages databases and ensures data is secure and available.

---

2. List 3 product companies and 3 service companies with one example of work each does.

Product Companies

1. **Microsoft** – Develops the Windows operating system and Microsoft Office.
2. **Google** – Creates products like Google Search, Gmail, and Android.
3. **Apple** – Designs iPhones, Macs, and the iOS operating system.

Service Companies

1. **Tata Consultancy Services (TCS)** – Provides software development and IT consulting services.
2. **Infosys** – Offers digital transformation and software solutions for businesses.
3. **Wipro** – Delivers IT consulting, cloud, and business process services.

---

3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Term	Full Form	Use
GitHub	Git + Hub (Git-based code hosting platform)	Used to store, manage, and collaborate on source code.

Term	Full Form	Use
Slack	*Not an acronym*	Used for team communication and collaboration.
Jira	*Not an acronym*	Used to track projects, tasks, and software bugs.
IDE	Integrated Development Environment	Used to write, edit, compile, and debug code.
API	Application Programming Interface	Allows different software applications to communicate with each other.

---

4. Name any 5 programming languages and mention one popular use case for each.

Programming Language	Popular Use Case
C	Operating systems and embedded systems
C++	Game development
Java	Android application development
Python	Data science and artificial intelligence
JavaScript	Interactive websites and web applications

---

5. What is the difference between a developer and a tester? (2 lines)

- A **developer** writes and develops software applications based on requirements.
- A **tester** checks the software for bugs and ensures it works correctly before release.

Session 2 –

1. Three Main Layers of Software Architecture

- **Presentation Layer (UI): The part of the application that users interact with, such as screens, buttons, and forms.**

- **Business Logic Layer: Processes data, applies business rules, and handles the application's main functionality.**
- **Data Layer (Database): Stores, retrieves, and manages application data in a database.**

---

2. Define Client and Server (with Example)

- **Client: A client is a device or application that sends a request for a service or data.**
- **Server: A server is a computer or system that receives the request, processes it, and sends back the required data.**

Real-life Example:
When you open YouTube on your phone, your phone is the client, and YouTube's computers that send videos to your phone are the server.

---

3. Two Differences Between Desktop Apps, Web Apps, and Mobile Apps

Desktop Apps	Web Apps	Mobile Apps
Installed on a computer.	Run in a web browser.	Installed on a smartphone or tablet.
Can work without the internet (many apps).	Usually require an internet connection.	Can use phone features like GPS and camera.

---

4. Three Databases and Their Types

Database	Type
MySQL	SQL
PostgreSQL	SQL
MongoDB	NoSQL

Session 3 –

1. Full Forms and One-Line Use

Term	Full Form	One-Line Use
HTTP	HyperText Transfer Protocol	Used to transfer web pages between a browser and a web server.
HTTPS	HyperText Transfer Protocol Secure	Used to securely transfer web pages using encryption (SSL/TLS).
FTP	File Transfer Protocol	Used to upload and download files between computers over a network.
SMTP	Simple Mail Transfer Protocol	Used to send email from one mail server to another.

---

2. Difference Between IPv4 and IPv6 (2 lines)

- **IPv4** uses **32-bit** addresses (e.g., 192.168.1.1) and supports about **4.3 billion** unique addresses.
- **IPv6** uses **128-bit** addresses (e.g., 2001:db8::1) and supports a much larger number of unique addresses.

---

3. nslookup google.com IP Address

I can't run nslookup on your computer, so I can't provide the exact IP address you would get.

Run this command in **Command Prompt** (Windows) or **Terminal** (Linux/macOS):

nslookup google.com

Example output (may vary depending on your location and time):

Name: google.com
Address: 142.250.190.14

Note: Google uses multiple IP addresses, so your result may be different.

---

4. HTTP Status Codes

Status Code	Meaning
200 OK	The request was successful.
301 Moved Permanently	The requested page has permanently moved to a new URL.
403 Forbidden	The server understands the request but denies access.
404 Not Found	The requested page or resource could not be found.
500 Internal Server Error	The server encountered an unexpected error.

---

5. DNS Hierarchy with www.google.com

- **Root (.)** → The top level of the DNS hierarchy (usually hidden).
- **TLD (Top-Level Domain)** → **.com**
- **Domain** → **google**
- **Subdomain** → **www**

Example:

```
Root (.)
  |
.com     ← Top-Level Domain (TLD)
  |
google   ← Domain
  |
www      ← Subdomain
```

Session 4 –

1. Six Phases of SDLC (Software Development Life Cycle)

Phase	Description
1. Requirement Analysis	Gather and understand the customer's requirements.
2. Design	Create the system architecture, database, and software design.
3. Development (Implementation)	Write the code based on the design.
4. Testing	Test the software to find and fix bugs.
5. Deployment	Release the software for users to use.
6. Maintenance	Update, improve, and fix issues after deployment.

---

2. Three Differences Between Agile and Waterfall

Agile	Waterfall
Development is done in small iterations (sprints).	Development follows a fixed sequence of phases.
Requirements can change during the project.	Requirements are fixed before development starts.
Customer feedback is taken regularly.	Customer feedback is usually taken after the final product is completed.

---

3. Three Main Scrum Roles

Role	Responsibility
Product Owner	Defines product requirements and prioritizes the backlog.
Scrum Master	Helps the team follow Scrum practices and removes obstacles.
Development Team	Designs, develops, tests, and delivers the product increment.

---

4. What is a Sprint in Scrum?

A Sprint is a fixed period during which the Scrum team completes a set of planned tasks and delivers a working product increment.

Typical Duration: 1–4 weeks (most commonly 2 weeks).

---

5. Jira Board Columns

Column	Meaning
Backlog	A list of all pending tasks that have not yet been planned.
To Do	Tasks that are ready to be started.
In Progress	Tasks that are currently being worked on.
Done	Tasks that have been completed successfully.

Session 5 –

1. Git Commands

Task	Git Command
Initialize a repository	git init
Stage all files	git add .
Commit with a message	git commit -m "Initial commit"
Push to remote	git push origin main
Pull latest changes	git pull origin main

---

2. Difference Between git fetch and git pull (2 lines)

- **git fetch downloads the latest changes from the remote repository but does not merge them into your current branch.**

- **git pull downloads the latest changes and automatically merges them into your current branch.**

---

3. Create a GitHub Repository and Share the Link

I can't create a GitHub repository or push files from your computer.

After creating and pushing your repository, share the link in this format:

https://github.com/your-username/repository-name

Example:

https://github.com/johndoe/git-practice

---

4. Difference Between a Fork and a Pull Request (2 lines)

- **Fork: A copy of another user's repository in your own GitHub account, allowing you to make changes independently.**
- **Pull Request (PR): A request to merge your changes from a branch or fork into another repository.**

---

5. Commands to Create a Branch, Commit, and Merge

```
# Create and switch to a new branch
git checkout -b feature-test

# Make changes to a file
echo "This is a test." >> test.txt

# Stage the changes
git add test.txt

# Commit the changes
git commit -m "Added test file"

# Switch back to the main branch
git checkout main
```

```
# Merge the feature-test branch into main
git merge feature-test
```